

Curriculum for Undergraduate Degree (B.Tech.) in Automobile Engineering (w.e.f. AY: 2020-21)

Part III: Detailed Curriculum

Fourth Semester

Course Name:	Numerical Methods		
Course Code:	BS-M404	Category:	BS
Semester:	Fourth	Credit:	2
L-T-P:	2-0-0	Pre-Requisites:	Some concepts from basic math – algebra, geometry, pre-calculus and statistics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To compute different numerical errors in computations.		
2	To learn interpolation techniques.		
3	To apply the techniques for solving integrations, ODEs.		
4	Solve linear and non-linear equations.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Approximation in numerical computation: Truncation and rounding errors, Fixed and floating-point arithmetic, Propagation of errors.	2
2	Interpolation: Newton's Forward Interpolation, Newton's Backward Interpolation, Lagrange's Interpolation, Newton's Divided Difference Interpolation	4
3	Numerical integration: General Quadrature Formula, Trapezoidal Rule, Simpson's 1/3 Rule, Expression for corresponding error terms	3
4	Numerical solution of a system of linear equations: Gauss Elimination Method, Matrix Inversion, LU Factorization Method, Gauss-Seidel Iterative Method	6
5	Numerical solution of Non-Linear equation: Bisection Method, Regula-Falsi Method, Newton-Raphson Method (4L)	4

6	Numerical solution of ordinary differential equation: Euler's Method, Runge-Kutta Methods, Predictor-Corrector Methods, Finite Difference Method	5
7	Measure of Central Tendency and Dispersion: Mean, median, mode and S.D.	3
8	Curve Fitting by Method of Least Square: Fitting a straight line of the form $y = a + bx$, Fitting a curve of the form $y = ax + bx^2$, $y = ab^x$, $y = ae^{bx}$, $y = ax^b$.	3
		30

Course Outcomes:

After completion of the course, students will be able to:

1	Demonstrate understanding of common numerical methods and how they are used to obtain approximate solutions to otherwise intractable mathematical problems.
2	Apply numerical methods to obtain approximate solutions to mathematical problems.
3	Derive numerical methods for various mathematical operations and tasks, such as interpolation, differentiation, integration, the solution of linear and nonlinear equations, and the solution of differential equations.
4	Analyse and evaluate the accuracy of common numerical methods.

Learning Resources:

1	C. Xavier: C Language and Numerical Methods.
2	A. K. Jalan and Utpal Sarkar, Numerical Methods-A Programming Based Approach, Orient Blackswan Private Ltd.
3	Dutta & Jana: Introductory Numerical Analysis.
4	J.B.Scarborough: Numerical Mathematical Analysis.
5	Jain, Iyengar, & Jain: Numerical Methods (Problems and Solution).
6	Balagurusamy: Numerical Methods, Scitech.
7	Baburam: Numerical Methods, Pearson Education.
8	N. Dutta: Computer Programming & Numerical Analysis, Universities Press.
9	Soumen Guha & Rajesh Srivastava: Numerical Methods, OUP.

10	Srimanta Pal: Numerical Methods, OUP.
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Course Name:	Materials Engineering		
Course Code:	ES-AUE401	Category:	Engineering Science Courses
Semester:	Fourth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	No-prerequisite
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	Understanding of the correlation between the internal structure of materials, their mechanical properties and various methods to quantify their mechanical integrity and failure criteria.		
2	To provide a detailed interpretation of equilibrium phase diagrams.		
3	Learning about different phases and heat treatment methods to tailor the properties of materials.		
4	Learning about different materials to apply engineering application.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Crystal Structure: unit cells, metallic crystal system, imperfection in solids (0D, 1D, 2D and 3D defects), Young's modulus, relations between true and engineering stress-strain curves, generalized Hooke's law, yielding and yield strength, hardness, ductility, resilience, toughness and elastic recovery.	5
2	Mechanisms of plastic and elastic deformations: slip and twinning, recover, recrystallization and grain growth, strengthening mechanism (Strain hardening, Precipitation hardening, Refinement of Grain, solid solution strengthening), toughening mechanism (process and bridge zone). Fracture: ductile and brittle fracture- Griffith's theory, Creep: mechanisms of Creep, Creep resistant materials, Fatigue: SN curve, factors affecting fatigue life, prevention of fatigue failure. Corrosion: mechanism, classification, causes, prevention.	5

3	Alloys: substitutional and interstitial solid solutions, Phase diagrams: interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron Iron-carbide phase diagram and microstructural aspects of ledeburite, austenite, ferrite and cementite, cast iron etc.	6
4	Matellic material: Steel, Cast Iron, heat treatment of Steel: Annealing, tempering, normalizing, hardening and spheroidising, isothermal transformation diagrams for Fe-C alloys and microstructure development. TTT-diagram and CCT curves, austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening. Ceramic Materials: What is ceramics; common ceramic materials and their characteristics; How ceramics are made—sintering and vitrification process; Ceramic structures; Properties and applications. Polymer materials: What is polymer material; classification and properties (Visco-elastic) of polymer materials and their application; manufacturing process; polymer structures.	8
5	Composite materials: What is composites; concept about fiber and matrix, classification, polymers matrix composite, metal matrix and ceramic matrix composites and their applications; How composites are made. Advanced materials and carbon based nano composite: Advanced steels (TRIP, TWIP, IF), Advanced polymer (UHMWPE), advanced ceramic (UHTC), Advanced composite (hybrid composite), Graphene based nanocomposite etc.	6

6	Characterization of engineering materials (mechanical properties): Introduction to some simulation methods (Finite element and Molecular dynamics simulation), Introduction to some experimental methods (tensile, compression, bending and torsion tests). Application of engineering materials: Criteria for selecting materials for automotive components viz. cylinder block, cylinder head, piston, piston ring, gudgeon pin, connecting rod, crank shaft, cam shaft, cam, engine valve, gear, crown wheel and pinion, clutch plate, axle shaft, chassis, spring, body panel, brake lining, battery materials etc.	6
Total		36 L

Course Outcomes:

After completion of the course, students will be able to:

1	Student will be able to identify crystal structures for various materials and understand the defects in such structures.
2	Understand how to tailor material properties of different engineering materials.
3	Apply basic knowledge of materials structure -property to solve real-world problems
4	How to quantify mechanical integrity and failure in materials.

Learning Resources:

1	W. D. Callister, 2006, "Materials Science and Engineering-An Introduction", 6 th Edition, Wiley India.
2	V. Raghavan, 2004 "Material Science and Engineering", 5 th Edition, Prentice Hall of India Private Limited.
3	R. K. Rajput, 2010 "Fundamentals Of Material Science" 2 nd Edition, S. K. Kataria & Sons.
4	Smith William, 2008 "Material Science and Engineering" 4 th Edition, Tata McGraw Hill Education Private Limited.
5	Ever J. Barbero, 2013 "Finite Element Analysis of Composite Materials with Abaqus", 1 st Edition, CRC Press.

Course Name:	Strength of Materials		
Course Code:	PC-AUE401	Category:	PC
Semester:	Fourth	Credit:	4
L-T-P:	3-1-0	Pre-Requisites:	Knowledge of ES-AUE301 (Engineering Mechanics.)
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To know the behavior of material at various loading conditions in compression and tension. Analysis and recognize of principle stresses.		
2	Understand and analyze shear force and bending moment in various loading conditions.		
3	To know the phenomenon of bending and deflection of different sections and its analysis.		
4	To understands various columns sections and geometrical analysis.		
5	Concepts of strain energy, thin cylinder, torsion and numerical analysis		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Deformation in solids - Hooke's law, stress and strain- tension, compression and shear stresses – elastic constants and their relations- volumetric, linear and shear strains - principal stresses and principal planes - Mohr's circle.	12
2	Beams and types of transverse loading on beams- shear force and bend moment diagrams - Types of beam supports, simply supported and over-hanging beams, cantilevers. Theory of bending of beams, bending stress distribution and neutral axis, shear stress distribution, point and distributed loads.	10
3	Deflection of a beam using double integration method, computation of slopes and deflection in beams, Maxwell's reciprocal theorems. Buckling of columns, Euler's theory, critical loads for different types of constraints.	10
4	Torsion, stresses and deformation in circular and hollow shafts, stepped shafts, deflection of shafts fixed at both ends, stresses and deflection of helical springs.	8
5	Axial and hoop stresses in cylinders subjected to internal pressure, deformation of thick and thin cylinders, deformation in spherical shells subjected to internal pressure.	8
Total		48



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Course Outcomes:

After completion of the course, students will be able to:

1	Explain stress, strain, establish relationship between them and apply concepts of stress, strain to solve numerical problems.
2	Compute Shear Force and Bending Moment for determinate beams and draw Shear Force and draw Bending Moment Diagrams for various loading conditions.
3	Determine various stresses and draw stress diagrams using the knowledge of bending and shear concepts. Ability to solve numerical problems of deflection.
4	Explain theory of column failure with different support conditions, and develop numerical ability to solve numerical problems.
5	Solve numerical problems by applying knowledge of strain energy, torsion and thin cylinders and spherical shells.

Learning Resources:

1	Rajput.R K, "Strength of Materials", S.Chand& Co Ltd, New Delhi, 6th Edition, 1January 2015,
2	Bedi D.S., "Strength of Materials", Khanna Publishing House, Sixth edition ,1 January 2019.
3	Jindal U C, "Strength of Materials", Asian Books Pvt Ltd, New Delhi, Second edition 2013.
4	Subramanian R, "Strength of Materials", Oxford University Press, Third edition 15 June -2016.
5	Bansal R.K, "Strength of Materials", Lakshmi Publications(P)Ltd, Sixth edition 1 January 2018
6	Ferdinand P Been, Russell Johnson,J.R. and John J Dewole, "Mechanics of Materials", Tata Mcgraw Hill Publishing Co Ltd, New Delhi, 2005. 3.
7	Alternative NPTEL/SWAYAM Course: PROF. SRIMAN KUMAR BHATTACHARYYA

Course Name:	Machine Tools and Machining Technology		
Course Code:	PC-AUE402	Category:	PC
Semester:	Fourth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	NIL
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To impart knowledge and train students with the basic principles, applications and, parametric relationships in area of metal cutting and machining (conventional, un-conventional and NC/CNC) processes		
2	To develop capability among students for analyzing, problem solving and part programming in the area of metal cutting and machining (conventional, un-		

	conventional and NC/CNC) processes.
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Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Metal Cutting: Tool geometry and single point cutting tools, orthogonal and oblique cutting, rake, cutting tool signature; Chip shape and chip formation, chip tool interface, chip flow, built up edge, machined surface. Forces during turning, Merchant's circle diagram for cutting forces, force systems at chip tool interface and shear plane, velocity relationships and problems. Tool wear and tool life, Surface finish and integrity, Machinability, cutting tool materials, Cutting fluids, Coating	9
2	Conventional Machining processes: General principles of working and commonly performed operations in the following machines: Lathe, Shaper, Planer, milling, drilling and grinding machine. Estimation of machining time and Material removal rate.	8
3	Unconventional Machining Processes: Principles, process parameters and applications of Abrasive Jet Machining, Ultrasonic Machining, Electrical Discharge Machining, wire EDM; Electro-chemical machining, Laser Beam Machining, Plasma Arc machining, Water jet machining, Electron Beam Machining.	6
4	NC/CNC Machine Tools and Systems: Introduction to Numerical control, Advantages and applications, CNC (open loop and closed loop), DNC and Adaptive control. Components of NC /CNC machine tool, working principle of CNC lathe, milling machine, machining center. Manual Part Programming using ISO G and M Codes in CNC lathe and milling machine for simple jobs.	9
5	Principles of work holding: Jigs and fixtures, principles, applications and design.	4
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Describe construction, working principles & applications of conventional, non-conventional and numerically controlled machine tools and single/multi point cutting tools.	
2	Interpret relationship between different parameters and forces associated with cutting tools and machining operations.	
3	Solve problems related to parameters involved in metal cutting and machining operations	

4	Develop manual part programming for simple jobs using CNC lathe and milling machines
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Learning Resources:

1	Mikell P. Groover, Fundamentals of Modern Manufacturing: Materials, Processes, and Systems, Wiley India Publication, 2018.
2	Kalpakjian and Schmid, Manufacturing Processes for Engineering Materials (5th Edition)- Pearson India, 2014.
3	P. N. Rao, Manufacturing Technology – Metal Cutting and Machine Tools, Volume2 (5th Edition), Tata McGraw Hill Pub, 2018.
4	A.B. Chattopadhyay, Machining and Machine Tools (2 nd Edition) Wiley Publications, 2017
5	A. Ghosh & A.K. Mallick, Manufacturing Science, (2 nd Edition) East west Press Pvt. Ltd, 2010.
6	M.P. Groover, Automation, Production Systems and Computer Integrated Manufacturing, Pearson Education, 2016
7	Y. Koren, Computer Control of Manufacturing Systems, McGraw Hill Education, 2017.
8	P.C. Sharma, A Textbook of Production Engineering, S. Chand Pub, 1999.

Course Name:	Theory of Machines		
Course Code:	PC-AUE403	Category:	PC
Semester:	Fourth	Credit:	4
L-T-P:	3-1-0	Pre-Requisites:	Engineering Mechanics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	To impart knowledge of the kinematics and dynamics of planar single degree freedom mechanisms
2	To develop skills for interpreting and analyzing linkages, cams, gears, flywheels and governors.
3	To enable students to do analysis for dynamic forces, balancing and vibration of mechanical systems

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Mechanisms: Terminology and definitions, degree of freedom, mobility. Grashoff's law. Mechanical advantage. Transmission angle. Description of common mechanisms, introduction to four bar spatial mechanisms.	6L

	Kinematic inversions – Four bar chain, slider crank chain, applications of mechanisms.	
2	Kinematics: Displacement, velocity and acceleration analysis of simple mechanisms, graphical method, velocity and acceleration polygons, instantaneous centre method. Velocity and acceleration analysis using loop closure equations, kinematic analysis of simple mechanisms- slider crank mechanism, coincident points Coriolis component of acceleration	5L+3T
3	Cams: Classification of cams and followers-Terminology and definitions- Displacement diagrams- Uniform velocity, simple harmonic constant acceleration & deceleration, and cycloidal motions- derivatives of follower motions	3L+2T
4	Gears: Spur gear terminology and definitions. Fundamental law of gearing and conjugate action, spur gear contact ratio and interference/ undercutting. Helical, bevel, worm, rack and pinion gears (basics only). Gear trains, epicyclic gear trains, differentials.	4L+2T
5	Force Analysis of Mechanism, flywheel and governors: Dynamic force analysis of slider crank mechanism. Piston and crank effort, inertia, torque, Turning moment diagrams, fluctuation of energy, fly wheel. Governors: Use and classification; Study and analysis of Porter, Proell and Wilson-Hartnell governors; Sensitiveness, stability, isochronism, hunting, effort and power of governors.	5L+2T
6	Balancing: Balancing of Reciprocating and Rotating Masses- Static balancing, Unbalance of force or moment, Dynamic balancing of rotating masses- graphical and analytical methods; Swaying couple; Hammer blow	6L+2T
7	Vibration: Basic features of vibratory Systems-Elements, degrees of freedom, single degree of freedom system. Undamped free vibration, equation of motion, natural frequency. Damped free vibration, equation of motion, logarithmic decrement, critical speed of shaft.	5L+1T
Total		36L+12T
Course Outcomes:		
After completion of the course, students will be able to:		
1	Distinguish between various mechanisms along with their inversions.	
2	Explain and compute displacement, velocity and acceleration for various mechanisms and cam profiles for various followers.	
3	Distinguish and analyze working of various gears, gear trains, governors, flywheels and dynamic forces acting on mechanical systems.	



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4	Determine the balancing of masses for rotating and reciprocating machine elements
5	Interpret the various principles of vibrations of different systems

Learning Resources:

1	S S Rattan, "Theory of Machines", (4 th Edition) Tata McGraw -Hill Publishers, New Delhi, 2014.
2	R. L Norton, Kinematics and Dynamics of Machinery, McGraw Hill India, 2010.
3	Thomas Bevan, "The Theory of Machines", (3 rd edition) Pearson Education India, 2009.
5	John J. Uicker, Gordon R. Pennock & Late Joseph E. Shrigley, "Theory of Machines and Mechanisms", SI Edition, International Version, 2009
6	R.K. Bansal, J.S. Brar, "A Text book of Theory of Machines", (in SI units) (5 th Edition), 2016
7	Sadhu Singh, "Theory of Machines: Kinematics and Dynamics", (3 rd Edition), Pearson Education India, 2011.

Course Name:	Metrology and Instrumentation		
Course Code:	PC-AUE404	Category:	PC
Semester:	Fourth	Credit:	03
L-T-P:	3-0-0	Pre-Requisites:	No-prerequisite
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	To provide basic idea about the working principle and applications of devices of digital measurement system and transducers.
2	To provide basic idea about working principle and applications of some indicating and display devices and their applications.
3	To provide basic idea about working principle and applications of devices of force, torque, strain, stress and temperature measurement.
4	To understand the working of linear and angular measuring instruments.
5	To familiarize with the working of optical measuring instruments and fundamentals of limits and limit gauges

Course Contents:

Module No.	Description of Topic	Contact Hrs.
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1	Concept of measurement:-Introduction to Metrology; Need for high precision measurements; Terminologies in Measurement- Precision, accuracy, sensitivity, calibration, resolution. Errors in Measurement, types of errors, Abbe's Principle. Basic standards of length- Line standard, End standards, Wavelength standard; Various Shop floor standards. Linear Measurement – Slip gauges, wringing, grades; Surface plate; Dial indicators; Height gauges and Vernier calliper; screw gauge. Comparators-mechanical, electrical, optical and pneumatic. Angular Measurement – Bevel protractor; Sine Bar, principle and use of sine bar, sine centre; Angle gauges. Spirit level; Angle Dekkor; Clinometers.	7
2	Limits and Limit gauges – Making to suit, selective assembly, systems of limits and fits; Types of fits; Hole basis system and Shaft basis system. Tolerance, allowance and deviation (as per BIS). Limit Gauges – GO and NO GO gauges; types of limit gauges. Gauge design - Taylor's principle of gauging; Gauge tolerance, disposition of gauge tolerance, wear allowance. Optical Measuring Instruments: - Benefits of light waves as standards; Monochromatic light; Principle of Interference. Interference band, optical flat, surface measurement. Interferometers – NPL, Pitter-NPL, auto collimator.	7
3	Screw thread measurement – Screw thread terminology; Measurement of major diameter; root diameter; pitch; effective diameter with two wire method and three wire method. Measurement of flank angle and form by profile projector and microscope. Measurement of surface texture – roughness and waviness; Analysis of surface traces, peak to valley height, R.M.S. value, Centre Line Average and Ra value, Rt, Rz etc. Methods of measuring surface roughness – Stylus probe, Tomlinson surface meter, Talysurf; surface roughness measurement – assessment length, roughness width cut-off, sampling length and evaluation length.	7
4	Introduction to Digital Measurement – significance of Digital measurement; methods; Classification. Stages in generalized measuring system – Sensor-Transducer stage, Signal-Conditioning stage, Readout Recording stage; Types of input quantities. Definition of terms, calibration, standards, generalized measurement systems, static and dynamic performance characteristic of measuring devices. Drift, Resolution, Threshold, Hysteresis, Static calibration. Dynamic characteristics different order systems and their response, Measuring lag, Fidelity, Dynamic error.	5
5	Analysis of experimental data; knowledge of instruments: different indicating and display devices: PMMC, MI type instruments, Ammeter, Voltmeter, LED, LCD, 7- segment display type instruments. Concept of feedback; Open and close loop control systems. Instrumentation for measurement of position and displacement: Active and Passive transducers. Performance characteristic of different types of Transducers	5

	– Working, Classification of transducers. Motion and Dimension measurement – LVDT – Principle, applications, advantages and limitations.	
6.	Strain and Stress Measurement - Electrical resistance strain gauge - Principle, operation. Measurement of Force and Torque – Strain-Gauge Load Cells, Hydraulic and Pneumatic load cells . Force measurement using piezoelectric quartz crystal, Torque Measurement – Dynamometers – Mechanical ,Hydraulic and Electrical. Vibration measurement – Vibrometers and Accelerometers. Temperature Measurement – Use of Thermal Expansion – Liquid-in-glass thermometers, Bimetallic strip thermometer, Pressure thermometers. Thermocouples – Resistance Temperature Detectors (RTD); Thermistor; Pyrometers. Digital readouts, data acquisition and processing.	5
Total		36 L
Course Outcomes:		
After completion of the course, students will be able to:		
1	Get basic idea about working principle and applications of devices of digital measurement system and transducers.	
2	Get basic idea about working principle of some indicating and display devices and their applications.	
3	Get basic idea about working principle and applications of devices of force, torque, strain, stress and temperature measurement..	
4	Get basic idea about working principle and applications of devices of temperature measurement.	

Learning Resources:	
1	Anand K Bewoor, Vinay A Kulkarni, Metrology & Measurement, McGraw-Hill, 2009.
2	Ernest O. Doebelin, Dhanesh N. Manik, Measurement Systems Application and Design, McGraw-Hill, 2004.
3	Galyer J.F.W., Schotbolt C.R., Metrology for Engineers, ELBS,1990
4	Thomas G. Beckwith, John H. L., Roy D. M., Mechanical Measurements, 6/E, Pearson Prentice Hall, 2007.
5	R.K. Rajput, Mechanical Measurements & Instrumentation, S.K.Kataria & Sons.
6	A.K. Shawhany, Electrical n Electronic measurement System, Dhanpat Rai .

Course Name:	Manufacturing and Materials Testing lab		
Course Code:	PC-AUE491	Category:	Professional Core courses
Semester:	Fourth	Credit:	1.5
L-T-P:	0-0-3	Pre-Requisites:	No-prerequisite
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05
Course Objectives:			

1	To study the characteristics of material.
2	Interpret different engineering material properties
3	Identify different manufacturing technique
4	Translate suitable testing for proper application.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Impact tests: Charpy or Izod tests.	3
2	Fatigue test of a steel sample.	3
3	Experiments on friction: determination of coefficient of friction.	3
4	Determination of yield strength, ultimate strength, modulus of elasticity, percentage elongation and percentage reduction in areas.	3
5	Macro hardness test (Rockwell and Brinell).	3
6	Test for drawability of sheet metals through cupping test.	3
7	Heat treatment and microstructure characterization of steel	3
8	Sand preparation and testing: specimen preparation for testing permeability, clay content, grain fineness number, moisture content, green compression strength, green shear strength, splitting strength, hardness, etc.	3
9	Casting of metals after preparation of a suitable type moulds; Experiments on properties of post casting, fettling, cleaning, deburring, and polishing operations	3
10	Practicing smithy or forging of carbon steels and testing for its property changes;	3
11	Laboratory experiments in Fabrication processes to observe effects of varying process parameters in GMAW	3
12	Machining of Gear using milling machine	3
Total		36P

Course Outcomes:

After completion of the course, students will be able to:

1	Interpret different engineering material characters
2	Compute stress, strains and deformation of engineering materials.
3	Attain basic knowledge on pattern, Gear making.
4	Examine properties of mould material in casting.

Learning Resources:

1	Debabrata Nag and Abhijit Chanda, 2010, "Fundamentals of Strength of Materials", Wiley India.
2	W. D. Callister, 2006, "Materials Science and Engineering-An Introduction", 6 th Edition, Wiley India.
3	Kalpajian and Schmid, 2014, "Manufacturing processes for engineering materials", 7 th Edition, Pearson India.
4	Smith William, 2008, "Material Science and Engineering" 4 th Edition, Tata Mcgraw Hill Education Private Limited.
5	Mikell P. Groover, 2010, "Fundamentals of Modern Manufacturing: Materials, Processes, and Systems", 4 th Edition, Wiley Publication.

Course Name:	Environmental Science		
Course Code:	MC471	Category:	MC
Semester:	Fourth	Credit:	0
L-T-P:	2-0-0	Pre-Requisites:	Basic concepts of Environmental Science
Full Marks:	100		
Examination Scheme:	Semester Examination: 100		

Course Objectives:

1	Purpose: We as human being are not an entity separate from the environment around us rather we are a constituent seamlessly integrated and co-exist with the environment around us. We are not an entity so separate from the environment that we can think of mastering and controlling it rather we must understand that each and every action of ours reflects on the environment and vice versa. Ancient wisdom drawn from Vedas about environment and its sustenance reflects these ethos. There is a direct application of this wisdom even in
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	modern times.
2	Idea of an activity based course on environment protection is to sensitize the students on the above issues through following two type of activities.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	(a) Awareness Activities:	
	i) Small group meetings about any of the topic.	4
	ii) Slogan making event	2
	iii) Poster making event	5
	iv) Seminar on any of the topic.	4
	v) Preparation of a report on any of the topic regarding current scenario.	4
2	(b) Actual Activities:	
	i) Plantation	1
	ii) Gifting a tree to see its full growth	1
	iii) Cleanliness drive	1
	iv) Drive for segregation of waste	1
	v) Shutting down the fans and ACs of the campus for an hour or so	1
		24 L

Course Outcomes:	
After completion of the course, students will be able to:	
1	Explain basic concepts, man, society & environment, their interrelationship, mathematics of population growth and associated problems, steady state conservation system.



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2	Demonstrate natural environmental hazards like flood, earthquake, landslide-causes, effects and control/management.
3	Classify air pollution, water pollution, land pollution, noise pollution and their controls.
4	Study Elements of ecology and environmental management.

Learning Resources:

1	M.P. Poonia & S.C. Sharma, Environmental Studies, Khanna Publishing House, New Delhi, 2019
2	Environmental science by Gillbert G. Master